

A Transformer-Based Model for English to Yoruba Machine Translation

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Abstract— In Nigeria, Yoruba is a predominant language among over 500 spoken languages. Despite its importance, Yoruba is classified as a low-resourced language, lacking the extensive datasets and linguistic resources needed for advanced natural language processing (NLP) applications. While machine translation (MT) has advanced significantly over the past decade, these improvements mainly benefit high-resourced languages due to the large corpora required for training neural networks. Consequently, Yoruba remains under-resourced in computational linguistics, and NLP research for Yoruba is still in its early stages. This research focuses on developing a machine translation model for the agricultural domain, translating from English to the resource-scarce language Yoruba using transfer learning. The research involved creating approximately 11,000 agriculture-related texts and generating human translations for these texts. This translation pair was then used to train a model based on the pre-trained Google T5-small model. The transformer's ability to handle complex linguistic patterns and variable-length sequences improves translation quality and reduces training time and development complexity, even with a small training dataset. The model achieved a BLEU score of 0.5354 on the evaluation set against the BLEU score of 0.5369 obtained during the model training. Additionally, the translation quality was evaluated by humans in terms of representation accuracy, contextual correctness, and syntactic correctness. Using the mean opinion score of ten (10) local speakers, the translation quality of the model generated in this research ranks above translations from a state-of-the-art model in terms of syntactic and semantic correctness. The results demonstrated that the model developed in this research used domain-specific terms from the training corpus, providing more informative and beneficial translations for local farmers than Google Translate, which often relies on transliteration.

Keywords— Transformer-based, NLP, Yoruba, English, Machine Translation

I. INTRODUCTION

Natural Language Processing (NLP) is a rapidly growing field in Artificial Intelligence (AI) that enables seamless interaction between humans and machines, allowing computers to understand and generate meaningful responses in human languages [1]. NLP encompasses tasks such as text refinement, tokenization, sentiment analysis, and machine translation (MT), which involves translating texts between human languages using automated techniques [2], [3]. MT entails different approaches among is the Neural Machine Translation (NMT).

NMT advances MT by using neural networks for accurate translations but relies on large parallel datasets, making it more effective for high-resource languages than low-resource ones like Yorùbá [2], [4]. Transformers, introduced in 2017 are state-of-the-art deep neural networks utilizing an encoder-decoder architecture with self-attention mechanisms, enhancing performance in MT and other NLP tasks [5], [6].

Yorùbá, a tonal language with many dialects, is spoken by over 40 million people across West Africa, mainly in Nigeria, Benin, Togo, Sierra Leone, and Ghana [7]. As a low-resource Niger-Congo language, it requires extensive datasets for NLP applications to support international students and tourists [8]. It is one of Nigeria's three major indigenous languages, alongside Hausa and Igbo [9].

Many agricultural workers face literacy challenges due to socio-economic factors, limiting their access to crucial information. Since agricultural updates are often in English, Yorùbá speakers miss vital opportunities. Bridging this gap requires collaboration between computer scientists and linguists to develop NLP and MT solutions. This research focuses on a Transformer-based English-to-Yoruba MT model to enhance access to agricultural information and policies.

II. RELATED WORKS

Machine translation aims to bridge language barriers by enabling seamless text translation. Despite the challenges of linguistic ambiguity and complexity, it holds great potential for communication. While NMT has gained attention, transformer-based translation for agricultural texts from English to Yoruba remains a key area of interest.

Several methods have been proposed for English-to-Yoruba translation, each requiring specific resources. These include rule-based, statistical, neural, and example-based approaches. However, each method has limitations, highlighting the need for further research and development in this field.

The rule-based approach relies on linguistic rules for translation [10], [11]. The statistical approach analyzes bilingual corpora to learn word sequence probabilities [10], [12]. The example-based approach retrieves similar source texts and translations [10]. The neural approach uses deep learning for machine translation [11], [13].

To address issues with various methods identified, some researchers have translated from English to Yoruba, French to English, English to Malay to mention a few using the abovementioned approaches.

[1] A hybrid English-to-Yoruba translation system combining rule-based and statistical methods demonstrated higher fluency and intelligibility than previous models and Google Translate. Another study developed an RNN-based English-to-Yoruba translation model using a parallel Bible corpus, showing improved fluency and accuracy through encoder-decoder architecture [2], [14]. To address RNN limitations, a French-to-English Transformer-based model was implemented in PyTorch, enhancing efficiency and memory handling [15], [17].

Based on studies, research has been conducted on English-to-Yoruba translation. However, no studies have focused on a transformer-based model for English-to-Yoruba machine translation specifically within the agricultural domain. Therefore, it is important to conduct further research in this area, particularly in agriculture, where valuable information is often communicated in English. According to reviews, SMT, RBMT, EBMT are not sufficient enough for efficient translation on low resourced languages. In this research, transformer-based model was adopted for translating English to Yoruba, with emphasis on agricultural texts.

III. METHODOLOGY

The proposed Transformer-based model for English-to-Yoruba translation consists of four modules:

- i. **Data Curation Module:** It collects agricultural text and generates Yoruba translations to build the corpus for the proposed system.
- ii. **Corpus Preprocessing Module:** It entails the corpus cleaning, sentence segmentation, duplicate entry removal, and tokenization of the corpus.
- iii. **Model Training Module:** The corpus is partitioned into the training, validation, and testing sets, and the model is trained
- iv. **Evaluation Module:** It entails model evaluation, where the trained model is evaluated on the test set.

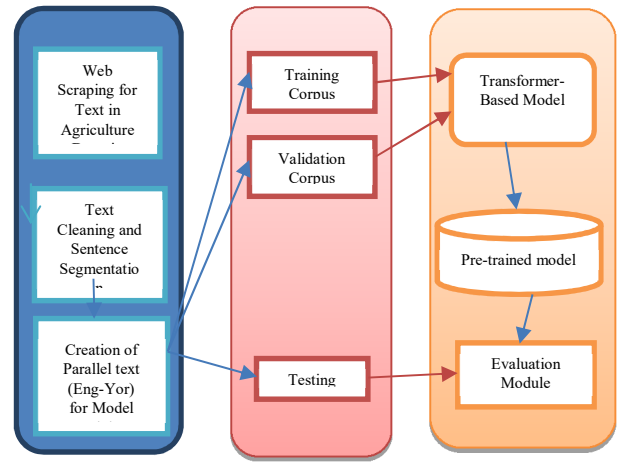


Fig. 1. Architecture of the Proposed System

A. Data Gathering and Cleaning

Initially, the SketchEngine tool was used to collect an agricultural text corpus using 578 verified keywords, but after processing, the corpus was reduced from 316,332 to 202,901 paragraphs and found unsuitable due to excessive research content. Extracting data from agricultural textbooks also proved insufficient. Given the importance of high-quality training data, the final approach leveraged a Large Language Model (LLM). ChatGPT 3.5 was used for dataset curation, with custom prompts refined to generate domain-specific text. Fig. 2 presents a sample prompt and the responses generated.

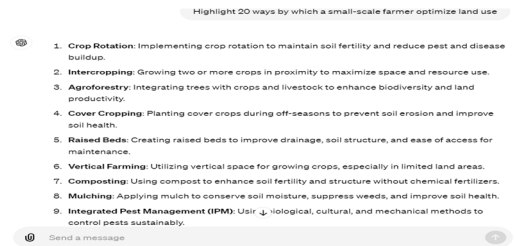


Fig. 2. Sample response from ChatGPT for a question prompt.

A total of 13,385 sentences were generated from the responses from ChatGPT.

Three major pre-processing tasks were carried out. This includes;

- i. The removal of the response number for each line,
- ii. The removal of duplicate entries after the merging of the files, and
- iii. The separation of punctuations from associated words.

This resulted in a Corpus with 11,053 sentences; the service of a linguist and local speakers was employed to generate the equivalent Yoruba translation for the Corpus.

The English and the corresponding Yoruba translations were aligned at the sentence level and placed in separate text files.

B. Model Training

The development of the machine translation model for translating agriculture-related text from English language into its Yoruba language equivalence in this research was based on the Transformer model introduced by [18] with Transfer learning.

Transfer learning has emerged as a potent method for training machine translation models on low resources language and small dataset requirements.

The transformer takes English text as input and outputs Yoruba words with the highest translation probabilities in Fig. 3. The encoder processes the input into an internal representation, while the decoder generates the predicted translation iteratively.

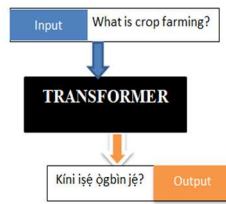


Fig. 3. Overview of the translation process

The original Transformer model, as shown in Fig. 4, can be partitioned into various stages. At the Encoder side of the architecture, the main task includes;

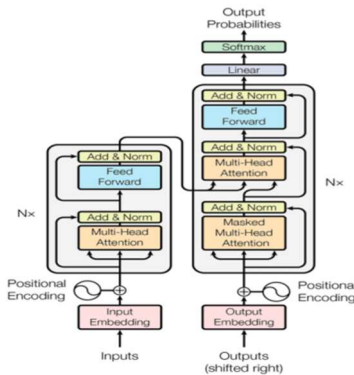


Fig. 4. Transformer model architecture

- **Input tokenization:** Tokenization occurs in both the encoder and decoder during training, segmenting input text into tokens. For example, the sentence *Kíni iṣẹ̀ ògbìn jẹ́?* is tokenized as ['kíni,' , 'iṣẹ̀,' , 'ògbìn,' , 'jẹ́,' , '?'].
- **Word embedding:** The word embedding transforms each token in the corpus into a vector representation. This involves the mapping of the distinct word into an integer representation and then substituting the integer representation with a vector of size 512.
- **Positional Encoding (PE):** PE is added directly to the input embedding at the base of the Transformer model.
- **Encoder training:** The summed word and positional embeddings for each token serve as input to the

encoder's first layer. The Multi-Head Self-Attention (MHSA) layer captures different aspects of the input by computing attention scores from Query, Key, and Value matrices. These matrices are split into multiple heads, trained separately, and then merged into a 512×512 matrix, as shown in Equation 1.

$$\text{MultiHead}(Q, K, V) \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)W^O$$

Where, $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$ (1)

- **Decoder:** The decoder mirrors the encoder but includes an encoder-decoder attention layer between the Masked MHSA, MHSA, feed-forward, and output layers, which consist of a linear and softmax sub-layer.

This study adopted the Bilingual Evaluation Understudy (BLEU) metric to assess the translation quality of the trained model. BLEU measures similarity between model-generated and human-generated texts by comparing n-grams. To prevent short sentences from receiving high scores, the precision score is adjusted using a Brevity Penalty (BP), as detailed in Equations 2 and 3.

$$BLEU = BP \cdot \exp\left(\frac{1}{n} \sum_{i=1}^n \log p_i\right) \quad (2)$$

$$BP = \begin{cases} 1, & \text{if } c > r \\ e^{(1-r/c)}, & \text{if } c \leq r \end{cases} \quad (3)$$

In this research, Python was used for text cleaning, model training, and evaluation. As shown in Table 1, 7,737 sentences were used for training, while 3,316 were used for validation. The dataset was loaded into a Python data-frame for preprocessing, with sample sentences from the training corpus displayed in Fig. 5.

[illegible]

Fig. 5. Sample sentences from the training corpus

C. Experimental Setup And Training

In this research, the dataset was divided into a training set and an evaluation set using a ratio of 70-30%. The evaluation set was further divided into a testing set and a validation set using a ratio of 66.66%- 33.33% as presented in table 1.

TABLE I. EXPERIMENTAL SETUP FOR TRAINING AND EVALUATION

Training		Evaluation			
%	Sentence count	<i>Validation</i>		<i>Testing</i>	
		%	Sentence count	%	Sentence count
70	7,737	33.33	1105	66.66	2211

The model was initially trained for 10 epochs and later extended to 20 epochs to see if increasing the number of epochs will have a positive effect on the translation accuracy. Aside the time taken to load the dataset, dependencies, and the pre-trained model, the training time for each epoch is approximately 5 minutes. Although human reference translations included full diacritics, they were removed before training due to text encoding inconsistencies. Analysis of the training corpus revealed variations in encoding for identical words, despite their visual similarity, as shown in Fig. 6.

1	ni-->ni,ni
2	afojusun-->afojusun,afojusun,afojusun
3	anfaani-->anfaani,anfaani,anfaani
4	asayan-->agayun,agayun
5	kan-->kan,kn,kn
6	ara-->ara,ara,ara
7	ise-->ise,ise,ise,ise,ise,ise,ise,ise,ise
8	irufe-->irufe,irufe,irufe,irufe
9	abuda-->abuda,abuda,abuda,abuda
10	ati-->ati,ati,ati,ati
11	eya-->aya,aya,aya
12	adojutofo-->adojutofo,adojutofo
13	ohun-->ohun,ohun,ohun
14	gbara-->gbara,gbara
15	le-->le,le,le,le,le
16	okunfa-->okunfa,okunfa,okunfa
17	bii-->bii,bii,bii,bii,bii
18	irugbin-->irugbin,irugbin,irugbin
19	bo-->bo,bo,bo,bo,bo
20	ti-->ti,ti,ti

Fig. 6. Selected base words and their variants from the training corpus

IV. DISCUSSION AND RESULT

The experimental results from training over ten epochs, along with the loss graph and BLEU score, are shown in Fig. 7 and Fig. 8. Both training and validation losses decreased over the first six epochs. However, Fig. 7 shows a slight increase in validation loss while training loss continued to decrease, with an improved BLEU score in Fig. 8. This indicates overfitting, where the model performs well on training data but struggles with unseen data.

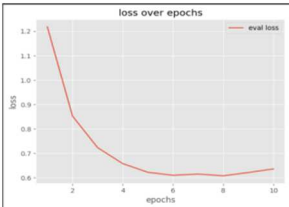


Fig. 7. Validation Loss graph over 10-epochs

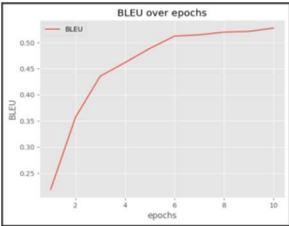


Fig. 8. BLEU Score graph over 10-epochs

To further analyze the model's behavior, training was extended by ten additional epochs, as shown in Fig. 9 and Fig. 10. While the BLEU score and training loss improved, the validation loss increased, indicating overfitting, as highlighted in Fig. 9. Fig. 10 shows significant accuracy gains between the first and sixth epochs, minor improvements between the sixth and sixteenth epochs, and no notable changes from the

sixteenth to the twentieth epoch. The model was configured to retain the best-performing version during training.

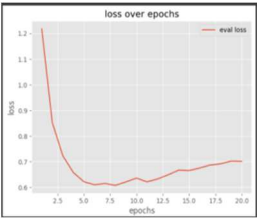


Fig. 9. Validation Loss graph over 20 epochs

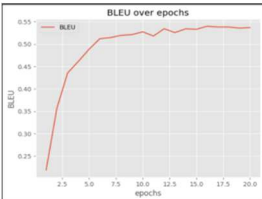


Fig. 10. BLEU score graph over 20-epochs

The model evaluation was conducted using translations from the test set. Table 2 summarizes the results, showing an average BLEU score of 0.5254, compared to the best BLEU score of 0.5369 achieved during training.

TABLE II. PRESENTATION OF THE EVALUATION RESULTS

Metric	Value
Loss	0.6245486736297607
BLEU	0.5254008281961307
Precisions	[0.83084388505572, 0.71491550330639, 0.64320452736758, 0.58838846561867]
Brevity Penalty	0.7630340288513032
Length Ratio	0.7871210312723801
Translation length	13189
Reference length	16756
Runtime (seconds)	63.0675
Sample seconds	14.905
Steps per seconds	1.871
Epoch	10.0

A. Discussion

For discussion, ten randomly selected English sentences were taken from the testing set. Table 3 displays one segment of these sentences along with their system-generated translations, reference translations, and Google Translations.

TABLE III. SAMPLE TRANSLATIONS FROM THE MODEL

SN	Source (English)	System output	Human Translation	Google translation
1	crispr-based gene editing techniques enable precise modifications to livestock genomes, offering potential benefits for improving animal health,	abuda-adamo to je mo (crispr) a maa seranwo isayida abuda-adamo ohun-osin, eyi a maa seranwo anfaani fun alekun ilera eranko, ipese, ati	etc abuda-adamo to je mo crispr, eyi a maa satunyewo ohun-osin, pipese awon anfaani fun ilera eranko, ibisi, ati igbayegbadun.	Awon ilana sisatunse jiini ti o da lori crispr je ki awon iyipada konge si awon genomes eran-osin, fifunni awon anfani ti o poju fun

	productivity, and welfare.	igbayegbadun.		imudaras ilera eranko, iselopo, ati iranlowo.
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To further assess the model's performance, the Mean Opinion Score (MOS) evaluation was conducted on selected test sentences in Table 3. Ten Yorùbá native speakers participated, with 60% female and 40% male, recruited from Ekiti (8) and Osun (2) states. Responses were collected via Google Forms, ranking translations based on general, contextual, and syntactic correctness, as well as potential improvements in Fig. 11.

For human translation, 70% agreed it captured all source text information, while 30% disagreed. For system-generated translation, agreement dropped to 60%, with 40% believing some information was missing. Google Translate results showed 20% strongly disagreed, 10% disagreed, 20% were neutral, 20% agreed, and 30% strongly agreed.

English Sentence 1	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
crispr based gene editing techniques enable precise modifications to livestock genomes, offering potential benefits for improving animal health, productivity, and welfare.					
abuda-adamo to je mo (crispr) a maa serawo isayida abuda-adamo ohun-osi, eyi a maa serawo anfaani fun alekun ilera eranko, pese, ati igbayegbadun.					
The translation captures all the information in the source text					
The translation contains contextual errors					
The translation contains syntactic errors					
The translation needs improvement					

Fig. 11. Sample Translation Rating Form

The human evaluation summary shows that the developed model effectively used domain-specific terms from the training corpus, making translations more accurate and relevant for local farmers. In contrast, Google Translate relied on transliteration (e.g., *jiini* for *gene*) or borrowed foreign words (*silos*, *genomes*, *pigweed*, *agroforestry*). The developed model closely matched human translation in preserving source text information, while Google Translate performed poorly. Hence, results from the model developed in this research are more informative and beneficial to Agriculture than the results from Google translate.

CONCLUSION

The design and implementation of a domain-based English-to-Yorùbá machine translation system using the transformer-based approach with transfer learning is a promising step in advancing the field of natural language processing for Indigenous and low-resourced languages. This approach leverages a pre-trained transformer model to translate agricultural texts, bridging the language gap for Yoruba-speaking farmers. It effectively handles Yoruba's complex structure and supports variable-length input and output, enhancing flexibility.

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